Records of Exchangeable Permutations

A complete finite counting proof of TLMC 00000009361

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Abstract

The upper-record indicators of a positionally exchangeable random permutation are independent Bernoulli variables with parameters $1, 1/2, \dots, 1/n$. We prove the exact joint pattern count using a bijection of actual permutations. A Lean 4.31 formalization verifies the permutation sample space, its duplicate-free enumeration, record semantics and the counting identity.

1 Statement and conventions

The official conjecture describes "the records of exchangeable permutations" and says their distribution is independent Bernoulli. We give its standard precise meaning. Let $\mathcal{P}_n$ be all permutations of $\{0, \dots, n-1\}$, written chronologically as $p = (p_1, \dots, p_n)$. Define the upper-record indicator

$$R_k(p) = \mathbf{1}\{p_k > p_j \text{ for every } j < k\}, \qquad 1 \le k \le n.$$

The empty comparison makes $R_1 = 1$. Exchangeability here means invariance of the law under arbitrary permutations of the *positions*. This is not merely conjugation invariance when a permutation is viewed as a group element. We assume distinct entries; ties would change the claim.

Theorem 1 (Exact joint count)**.** *For every $n \ge 0$ and $b = (b_1, \dots, b_n) \in \{0,1\}^n$,*

$$\#\{p \in \mathcal{P}_n : R_k(p) = b_k \text{ for all } k\} = \prod_{k=1}^{n} \begin{cases} 1, & b_k = 1, \\ k-1, & b_k = 0. \end{cases} \tag{1}$$

The empty product is 1. Thus $\#\mathcal{P}_n = n!$, and for a positionally exchangeable permutation the indicators $R_1, \dots, R_n$ are independent, with $\Pr(R_k = 1) = 1/k$.

2 The permutation bijection

For $0 \le r \le n$, define the increasing map

$$s_r(x) = \begin{cases} x, & x < r, \\ x+1, & x \ge r. \end{cases}$$

It is a bijection from $\{0, \dots, n-1\}$ to $\{0, \dots, n\} \setminus \{r\}$. Its inverse on that set is

$$d_r(y) = \begin{cases} y, & y < r, \\ y-1, & y > r. \end{cases}$$

The following is a bijection, with no hidden multiplicities:

$$\mathcal{P}_n \times \{0, \dots, n\} \longrightarrow \mathcal{P}_{n+1}, \qquad (q, r) \longmapsto (s_r(q_1), \dots, s_r(q_n), r).$$

Indeed the output contains every label exactly once. Conversely, remove the last entry r of any permutation in $\mathcal{P}_{n+1}$ and apply d_r to every remaining entry. This recovers a unique $q \in \mathcal{P}_n$. The identities $d_r(s_r(x)) = x$ and $s_r(d_r(y)) = y$ for $y \neq r$ prove both inverse identities. Starting with the unique empty permutation, this also proves $\#\mathcal{P}_{n+1} = (n+1)\#\mathcal{P}_n$ and hence $n!$.

3 Records and the joint count

Strict monotonicity gives $s_r(x) < s_r(y)$ if and only if $x < y$. Therefore all the first n record indicators are preserved by the bijection. The last entry r is a record if and only if $r = n$: when $r = n$, every previous label is smaller; when $r < n$, the previous labels include n, which is larger.

Fix a record pattern of length n. Each realizing permutation consequently has exactly one extension whose last entry is a record, and exactly n extensions whose last entry is not a record. These extensions are disjoint and exhaust all permutations with the prescribed extended pattern, by the bijection above. If $C_n(b)$ is the left-hand side of (1), then

$$C_{n+1}(b, 1) = C_n(b), \qquad C_{n+1}(b, 0) = nC_n(b), \qquad C_0(()) = 1.$$

Induction proves (1). In particular a pattern beginning with 0 has count zero, as required; no separate positive-size assumption is used.

4 Exchangeability and probability

The position-permutation action on $\mathcal{P}_n$ is transitive: one can reorder any permutation into any other. Invariance under this action therefore assigns the same probability to all $n!$ elements. Since their probabilities sum to one, each has probability $1/n!$. Dividing (1) by $n! = \prod_{k=1}^{n} k$ gives

$$\Pr(R_1 = b_1, \ldots, R_n = b_n) = \prod_{k=1}^{n} \begin{cases} 1/k, & b_k = 1, \\ 1 - 1/k, & b_k = 0. \end{cases}$$

This equality for every Boolean pattern is precisely mutual independence and the claimed Bernoulli laws. The *number* of records is the sum $N_n = \sum_{k=1}^{n} R_k$, not a family of independent copies of itself. Its probability generating function is

$$\mathbb{E}[z^{N_n}] = \prod_{k=1}^{n} \left(1 - \frac{1}{k} + \frac{z}{k}\right).$$

At $n = 0$, the law is concentrated on $N_0 = 0$. For any other fixed ordered set of n distinct labels, increasing relabeling preserves all comparisons and yields the same result.

5 Formalization and trust boundary

The source uses only Lean 4.31.0 and its bundled Std library. `Permutations.lean` defines actual lists of natural-number labels. The theorem `perms_complete` identifies membership of its enumeration with `List.Perm (List.range n)`; `perms_nodup` rules out repeated sample points and `perms_length` proves its factorial size. The intermediate validity predicate requires both no repeated entries and the exact membership condition $x \in p \iff x < n$.

For convenient recursion the enumeration stores a permutation in reverse chronological order. This is only a storage convention: `RecordSemantics.lean` separately defines a left-to-right scan which compares each entry with every earlier entry. `recs_chronological` proves agreement after reversal. `recs_extend` proves preservation of all earlier records and the last-record criterion $r = n$. `RecordsCount.lean` proves `chronologicalPatternCount_exact` for the complete duplicate-free chronological enumeration. Its `independent_bernoulli_joint_law` also proves that the exact rational uniform probability is the Bernoulli product. These are statements about actual permutations, rather than an unconnected sequence of rank choices.

The finite counting statements, normalized rational product identity and record/bijection correspondence are kernel-checked. The implication from positional exchangeability to the uniform measure and the interpretation of the joint product as independence are the elementary arguments in the preceding section; no general probability-library formalization is claimed. No admitted proofs, custom conclusion axioms, native-decision shortcuts or kernel bypasses are used. The audit file records the theorem axiom footprints. A separate Python test directly enumerates chronological permutations for $n = 0, \ldots, 8$ and checks every Boolean pattern. These tests are supplemental, not assumptions of the Lean proof.

Source

Official TLMC conjecture 00000009361.

Source blob SHA: `3c1e3568c9aa81b24440aa69a23149d953e1b4d4`. The English and Chinese statement is preserved in the accompanying source file.